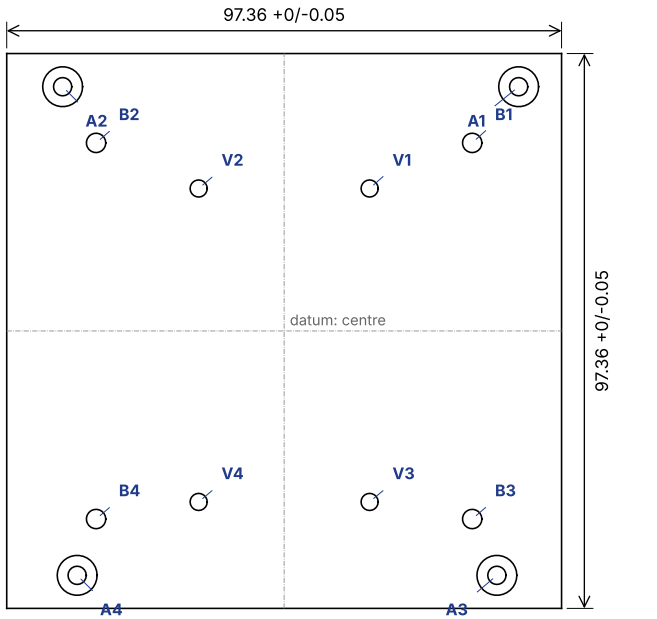
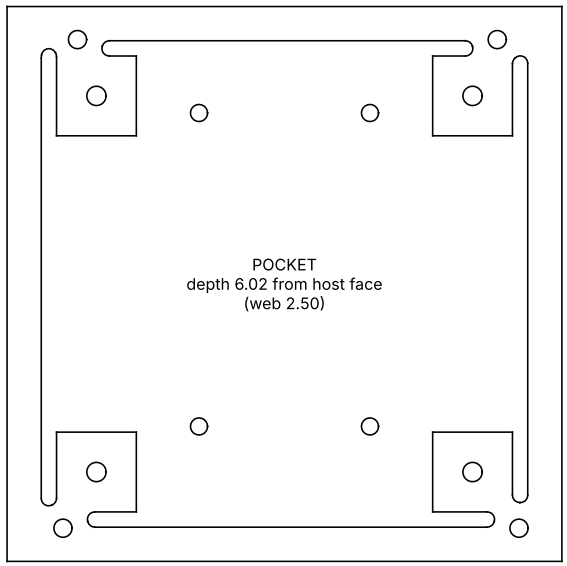
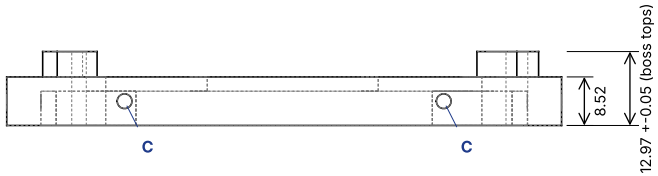
TOP VIEW (inner face)



BOTTOM VIEW (host face)





FRONT VIEW (from -Y), hidden lines dashed

FEATURE TABLE

A1 M3×0.5 INSERT thru boss and plate; boss dia 7.0, top 12.97 above host face.
X 41.15 Y 42.86 (+-0.05, PC/104 pattern)

A2 M3×0.5 INSERT thru boss and plate; boss dia 7.0, top 12.97 above host face.
X -38.87 Y 42.86 (+-0.05, PC/104 pattern)

A3 M3×0.5 INSERT thru boss and plate; boss dia 7.0, top 12.97 above host face.
X 37.33 Y -42.88 (+-0.05, PC/104 pattern)

A4 M3×0.5 INSERT thru boss and plate; boss dia 7.0, top 12.97 above host face.
X -36.32 Y -42.88 (+-0.05, PC/104 pattern)

B1 dia 3.40 thru (M3 host screw). X 33.00 Y 33.00

B2 dia 3.40 thru (M3 host screw). X -33.00 Y 33.00

B3 dia 3.40 thru (M3 host screw). X 33.00 Y -33.00

B4 dia 3.40 thru (M3 host screw). X -33.00 Y -33.00

V1 dia 3.00 thru web (vents the pocket). X 15.00 Y 25.00

V2 dia 3.00 thru web (vents the pocket). X -15.00 Y 25.00

V3 dia 3.00 thru web (vents the pocket). X 15.00 Y -30.00

V4 dia 3.00 thru web (vents the pocket). X -15.00 Y -30.00

C 8x M2.5×0.45 INSERT into the four edges, 2 per edge at +-28.0 from centre,
4.26 above host face, 6.5 deep (breaks into pocket)

P Underside pocket, depth 6.02, R3.0 corners. Leaves: perimeter band 6.0 wide; 4x pads 14 sq at B1-B4; 4x round pads dia 9.0 under A1-A4.

F Host face (bottom) flat within 0.05. Inner face flat within 0.05 (carries the converter thermal block).

R Pocket inside corners: where a round pad meets a square pad or the band, R1.30 (the shape a dia 2.6 cutter leaves; the 2.68 channels between pads and band stay open). Corners between the pocket walls and the web underside: R1.5 max.

NOTES

Units: millimetres. Third-angle projection. Tolerances unless stated: features and hole positions +-0.1; outline +-0.2.
Threads: ISO metric coarse, 6H. Holes marked INSERT are tapped for stainless free-running helical coil inserts, length 1.5D, supplied and installed by the shop; drill and tap to the insert maker's data. Inserts flush to 0.25 below.
Every drilled and tapped hole runs through to an open volume (venting). Break sharp edges 0.1-0.3; deburr all holes.
Finish: as machined for this unit. Please also quote: clear anodize MIL-A-8625 Type II Class 1, with the H01 host face and the panel/angle faces under rivets masked and conversion coated MIL-DTL-5541 Type II Class 3 (electrical bonding).
Assembly is by Carthage: fasteners locked and torqued per the SLANT assembly procedure (to be issued).
Fit-check unit built to the flight layout. Material certs appreciated; no inspection report required.
Model: the STEP file of the same part number governs any dimension not shown here.
Bottom face meets the host bracket; inner face carries the board stack and the converter block.

PART	**SLANT-H01 Base plate**
MATERIAL	6061-T6 plate, 5/8 in stock
QTY	1
MASS (model)	113 g
PROJECT	SLANT 1U housing, assembly rev 16
OWNER	Carthage College, Space Sciences
DRAWN / CHECKED	from model 2026-10-08 / _____
SHEET REV	D